

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[GH_FFCutsFits,CutFitFileGH] =
recover_selected_cuts('GH_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multicu
t_all\sel_cuts');
```

```
fn = fieldnames(GH_FFCutsFits);
%nf = 11
%for nf = 2:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei200_off110GHminFF';
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('*****\n');
```

```
*****
```

```
fprintf('Processing cuts: %s \n', selectCuts);
```

```
Processing cuts: Ei200_off110GHminFF
```

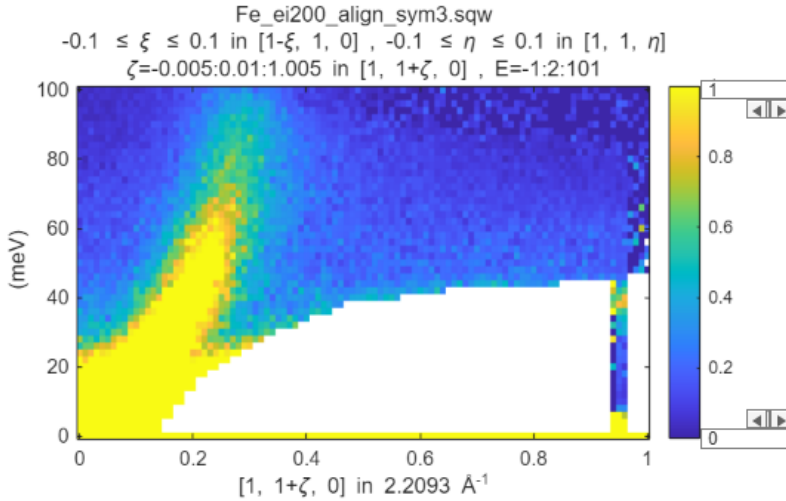
```
fprintf('*****\n');
```

```
*****
```

```
cuts_list = GH_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
```

```
cuts_list{j} = cuts_list{j}.set_instrument(mi);
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
```

end



```
dir_name = "GH";

dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

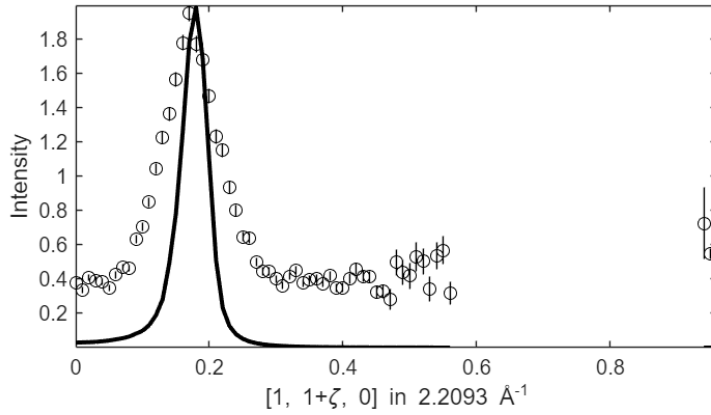
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GH_FFcutsFits.(selectCuts) = cuts_list;

GH_FFcutsFits = fit_multicutsFF_along_direction(...
    GH_FFcutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****
**** En = 35±5 going to 95
*****
```

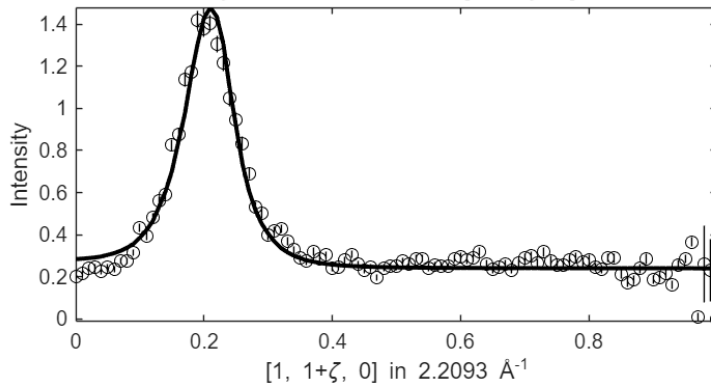
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $30 \leq E \leq 40$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



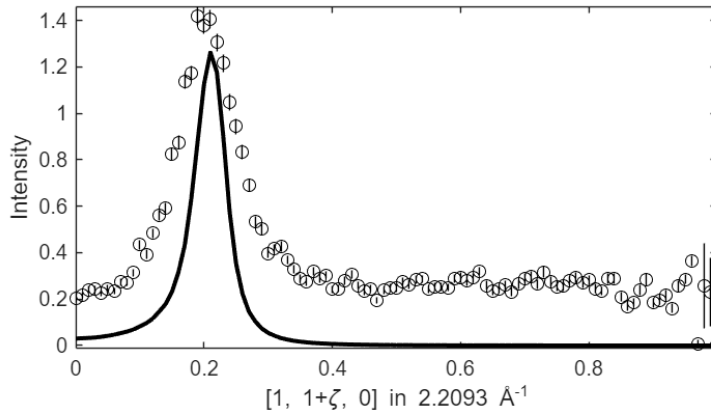
**** En = 45±5 going to 95

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Fe_ei200_align_sym3.sqw
 $S=0.81\pm0.021$; $J_0= 24\pm0.28$; $J_1= 0\pm 0$; $\gamma=7.62927\pm0.224815$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $40 \leq E \leq 50$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



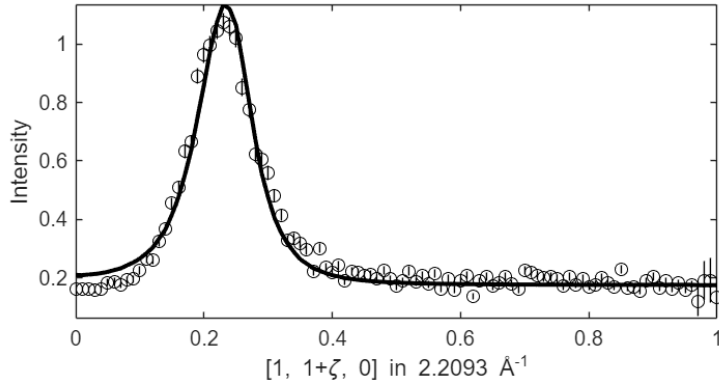
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $40 \leq E \leq 50$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



**** En = 55±5 going to 95

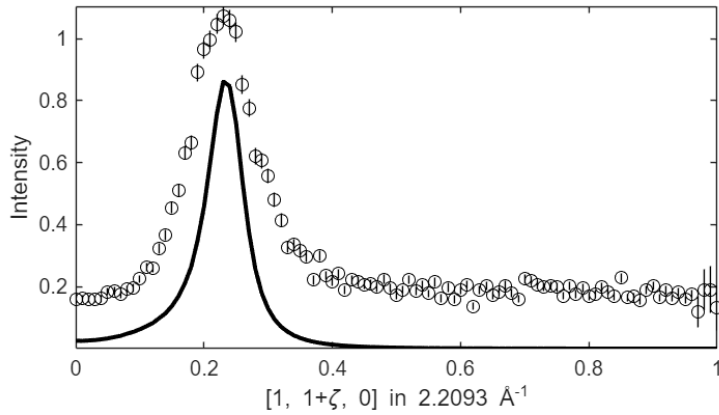
Fe_ei200_align_sym3.sqw

S=0.82±0.019; J0= 25±0.26; J1= 0± 0; gamma=12.1312±0.291065
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $50 \leq E \leq 60$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



Fe_ei200_align_sym3.sqw

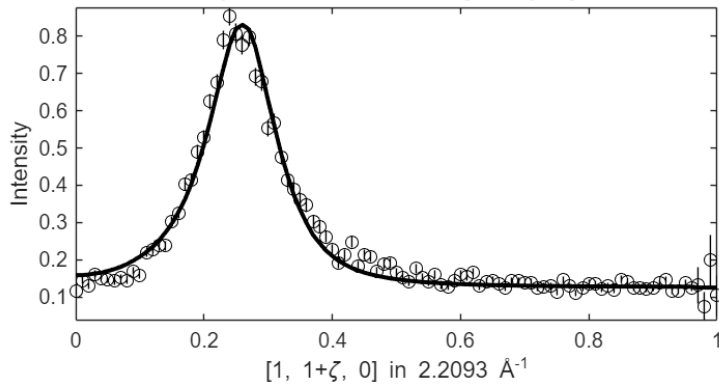
S=0.82±0.019; J0= 25±0.26; J1= 0± 0; gamma=12.1312±0.291065
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $50 \leq E \leq 60$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



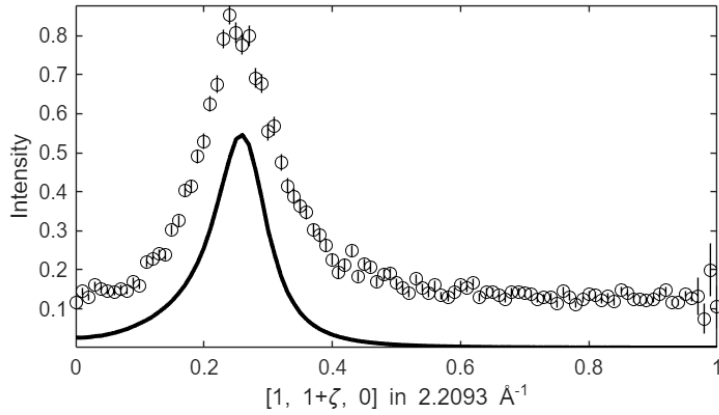
**** En = 65±5 going to 95

Fe_ei200_align_sym3.sqw

S=0.92±0.033; J0= 25± 0.3; J1= 0± 0; gamma=23.2166±1.4874
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $60 \leq E \leq 70$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$

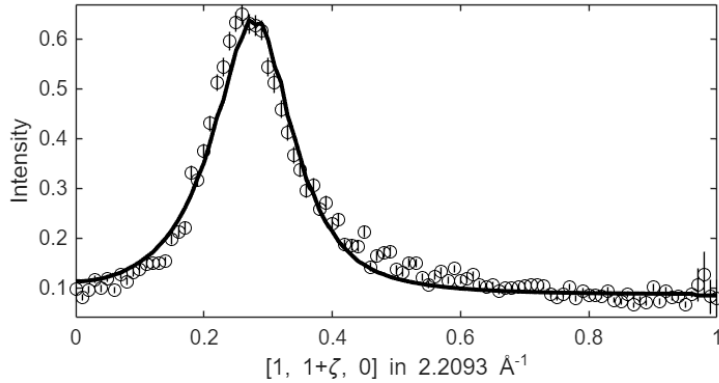


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $60 \leq E \leq 70$
 $\zeta = -0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$

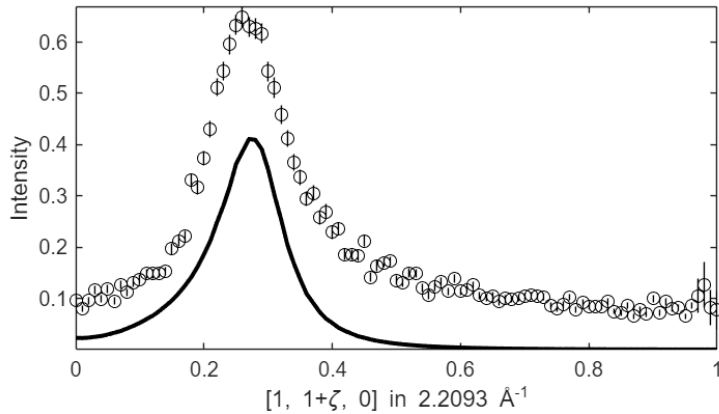


**** En = 75±5 going to 95

Fe_ei200_align_sym3.sqw
 $S=0.97\pm0.045$; $J_0=27\pm0.46$; $J_1=0\pm0$; $\gamma=32.395\pm2.55232$
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $70 \leq E \leq 80$
 $\zeta = -0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



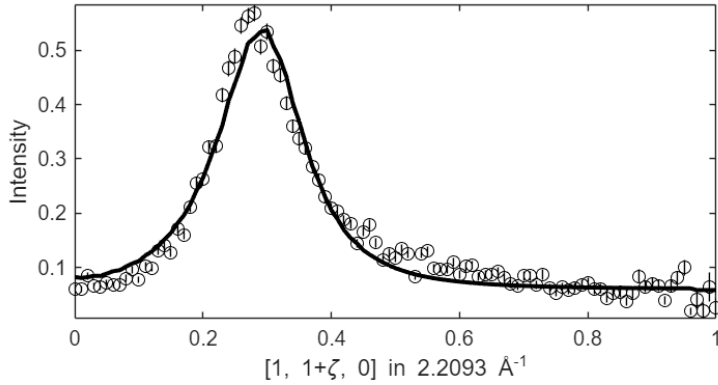
Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $70 \leq E \leq 80$
 $\zeta = -0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



**** En = 85±5 going to 95

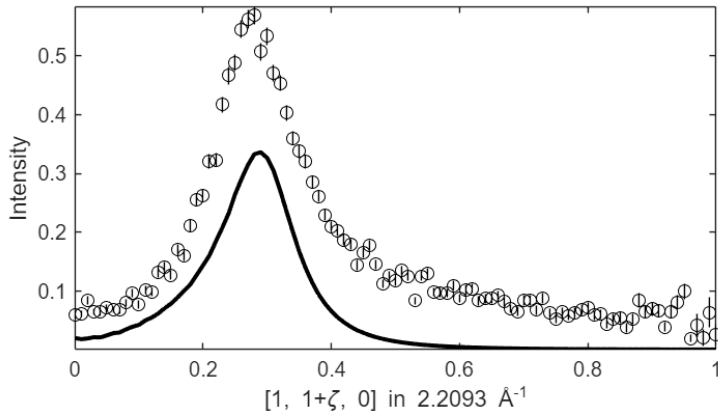
Fe_ei200_align_sym3.sqw

S=0.99±0.055; J0= 28±0.54; J1= 0± 0; gamma=40.2604±3.35635
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $80 \leq E \leq 90$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



Fe_ei200_align_sym3.sqw

S=0.99±0.055; J0= 28±0.54; J1= 0± 0; gamma=40.2604±3.35635
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $80 \leq E \leq 90$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$

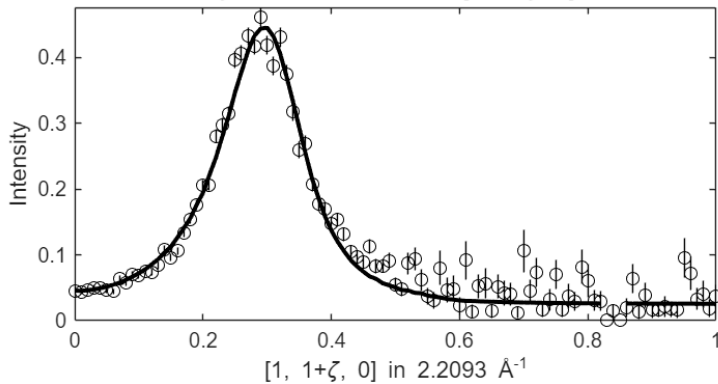


**** En = 95±5 going to 95

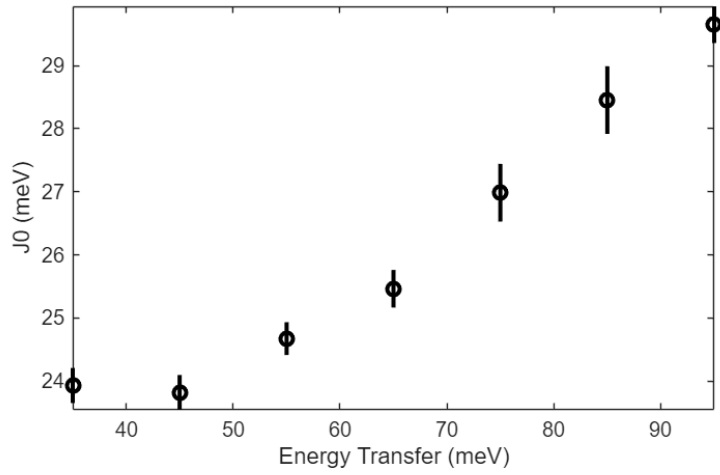
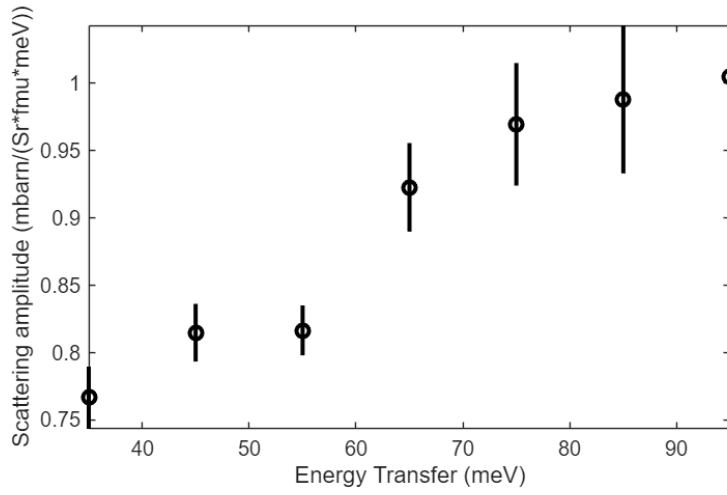
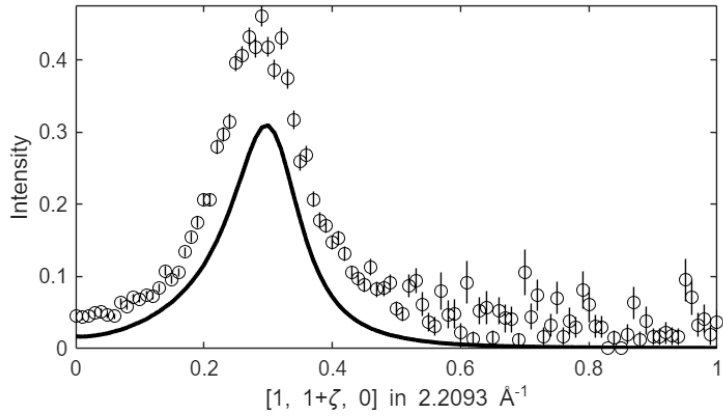
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802

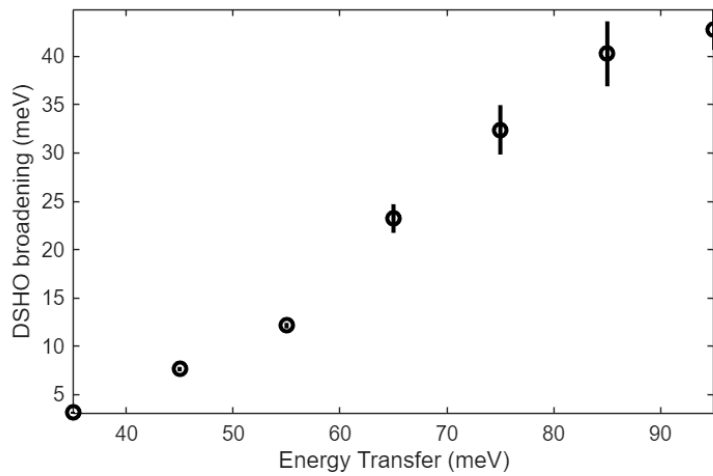
Fe_ei200_align_sym3.sqw

S= 1±0.036; J0= 30±0.29; J1= 0± 0; gamma=42.7626±2.09872
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $90 \leq E \leq 100$
 $\zeta=-0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$



Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1, 1, \eta]$, $90 \leq E \leq 10$
 $\zeta = -0.005:0.01:1.005$ in $[1, 1+\zeta, 0]$





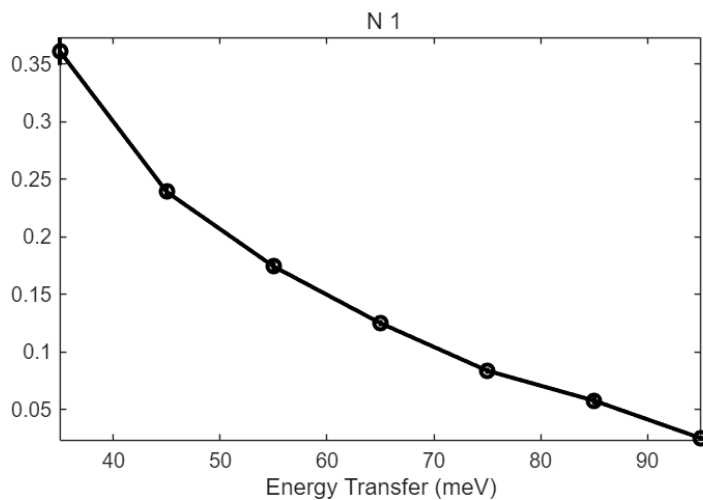
```

c1 = GH_FFCutsFits.(selectCuts);
GH_FFCutsFits.(selectCuts) = [c1{:}];

beep
beep
beep

all_fit_par = GH_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;

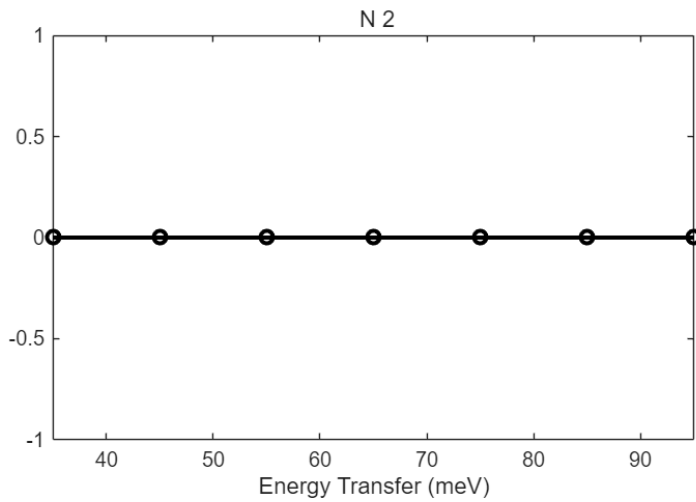
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```

```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GH.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GH_',selectCuts,'.pdf']);
export(thisFile,targ_file);
```

```
fit_src_struct = GH_FFCutsFits;
save(CutFitFileGH,"fit_src_struct",-v7.3');
close all;
%end
```